

Amer Al-Hiyasat

PhD Candidate, Department of Physics, MIT

hiyasat@mit.edu | ameralhiyasat.github.io

77 Massachusetts Avenue, Room 6C-431, Cambridge, MA 02139

EDUCATION

Massachusetts Institute of Technology 2022–2027 (expected)
PhD Candidate in Physics; advised by Julien Tailleur and Mehran Kardar; GPA: 5.0/5.0.

Yale University 2021
B.S. in Physics
B.S. in Molecular Biophysics and Biochemistry
GPA: 4.0/4.0, *summa cum laude*, with distinction in both majors.

HONORS AND AWARDS

- Kendall Teaching Fellow Award, MIT 2025
- Russell Henry Chittenden Prize — highest-ranking Yale graduate in the natural sciences 2021
- Pauli Murray Award, Yale 2021
- Sigma Pi Sigma Honor Society for Physics 2021
- Yale Science and Engineering Association Award 2020, 2021
- Phi Beta Kappa (elected junior year), Yale 2019

FELLOWSHIPS AND GRANTS

- Institute for Complex and Adaptive Matter (ICAM) Long-Term Exchange Award 2025, 2026
- APS Division of Biological Physics (DBio) Travel Award 2026
- Tushar Shah (1993) and Sara Zion (1994) Physics Fellowship, MIT 2025–2026
- MathWorks Science Fellowship, MIT 2023–2024
- Morton E. Goulder Presidential Fellowship, MIT 2022–2023
- Rosenfeld Science Fellowship, Yale 2019

PUBLICATIONS

Preprints

- [1] **A. Al-Hiyasat**. “Critical Properties and Glass Transitions in Randomly Coupled Fields”. (2026). arXiv: 2608.26279.

Published articles

- [2] **A. Al-Hiyasat**^{*}, D. W. Swartz^{*}, J. Gore, and M. Kardar. “Spatiotemporal noise stabilizes unbounded diversity in strongly-competitive communities”. *Proceedings of the National Academy of Sciences* (2026). To appear. arXiv: 2602.13423. ^{*}Equal contribution.
- [3] **A. Al-Hiyasat**, S. Ro, and J. Tailleur. “Cold tracer in a hot bath: In and out of equilibrium”. *Physical Review Letters* **136**, 117104 (2026).
- [4] **A. Al-Hiyasat**, S. Ro, and J. Tailleur. “Statistical mechanics of a cold tracer in a hot bath”. *Physical Review E* **113**, 034127 (2026).
- [5] Y. Tuna, **A. Al-Hiyasat**, A. D. Kashkanova, A. Dechant, E. Lutz, and V. Sandoghdar. “Electrostatic all-passive force clamping of charged nanoparticles”. *ACS Nano* **19**(10) (2025), 10173–10179.
- [6] **A. Al-Hiyasat**, Y. Tuna, Y. W. Kuo, and J. Howard. “Herding of proteins by the ends of shrinking polymers”. *Physical Review E* **107**, L042601 (2023).

- [7] R. Rai, K. Biju, W. Sun, T. Sodeinde, **A. Al-Hiyasat**, J. Morgan, X. Ye, X. Li, Y. Chen, and S. Chang. "Homology directed telomere clustering, ultrabright telomere formation and nuclear envelope rupture in cells lacking TRF2B and RAP1". *Nature Communications* **14**, 2144 (2023).
- [8] Y. Tuna, **A. Al-Hiyasat**, and J. Howard. "Simultaneous Interference Reflection and Total Internal Reflection Fluorescence Microscopy for Imaging Dynamic Microtubules and Associated Proteins". *Journal of Visualized Experiments* **183**, e63730 (2022).
- [9] A. Cicconi, R. Rai, X. Xiong, C. Broton, **A. Al-Hiyasat**, C. Hu, S. Dong, W. Sun, J. Garbarino, R. S. Bindra, C. Schildkraut, Y. Chen, and S. Chang. "Microcephalin 1/BRIT1-TRF2 interaction promotes telomere replication and repair, linking telomere dysfunction to primary microcephaly". *Nature Communications* **11**, 5861 (2020).

Manuscripts in preparation

- [10] **A. Al-Hiyasat**, D. G. Martin, Y. Kafri, M. Kardar, and J. Tailleur. "Generic Scale Invariance, Fluctuation-Induced Forces, and Long-Range Flows in Anisotropic Nonequilibrium Matter". Manuscript in preparation (2026).
- [11] D. Shah and **A. Al-Hiyasat**[†]. "Interfacial instabilities destroy order in a nonreciprocal lattice gas". Manuscript in preparation (2026). [†]Senior author.

RESEARCH EXPERIENCE

Department of Physics, Massachusetts Institute of Technology

Graduate student with Profs. Julien Tailleur and Mehran Kardar

2022–Present

Topic: Theory of nonequilibrium statistical mechanics.

- Studied the role of spatiotemporal noise in disordered generalized Lotka–Volterra ecological communities, revealing a novel resolution of the long-standing diversity–stability paradox (Ref. 2).
- Characterized the nonequilibrium dynamics of tracers in multi-temperature particle systems (Refs. 3 and 4).
- Characterized glass transitions in randomly coupled equilibrium field theories (independent project, Ref. 1).
- Studying the role of anisotropy in the critical dynamics of nonequilibrium particle systems (Ref. 10).
- Studying interfacial instabilities in a nonreciprocal lattice gas (independent project with undergraduate mentee Dhruv Shah, Ref. 11).

Department of Molecular Biophysics and Biochemistry, Yale University

Postgraduate associate

2021–2022

Undergraduate researcher with Prof. Jonathon Howard

2020–2021

Topic: Experimental and theoretical studies on the biophysics of microtubules.

Discovered a novel mechanism of protein localization to polymer ends (Refs. 6 and 8), using single-molecule experiments as well as analytical and numerical models. Co-developed a microfluidic system for high-speed imaging of microtubules and associated proteins. Mentored two undergraduates.

Department of Laboratory Medicine, Yale School of Medicine

Undergraduate researcher with Prof. Sandy Chang

2018–2020

Topic: Experimental studies on the molecular biology of telomeres.

Studied the role of the BRIT1-TRF2 interaction in the telomeric DNA damage response (Ref. 9). Characterized the "ultrabright telomeric foci" phenotype, uncovering a putative telomere clustering mechanism using chromosome-orientation FISH (Ref. 7). Developed a native telomere labeling system and a specialized image-analysis pipeline.

PRESENTATIONS

Invited and Selected Talks

- "Spatiotemporal Noise Stabilizes Unbounded Diversity in Strongly-Competitive Communities." Selected talk, *Non-equilibrium Dynamics, from Active Matter to Evolutionary Dynamics*, KITP June 2, 2026
- "Spatiotemporal Noise Stabilizes Diversity in Strongly-Competitive Metacommunities." Invited talk, *Harvard Condensed Matter Theory Kid's Seminar* Dec. 9, 2025
- "A Cold Tracer in a Hot Bath: In and Out of Equilibrium." Selected talk, *Recent Advances in Active Matter and Statistical Physics*, Leiden July 10, 2025

- “A Cold Tracer in a Hot Bath: In and Out of Equilibrium.” Selected talk, *MIT Biophysics Retreat* Sept. 30, 2024
- “Active Particles at Passive Phase Boundaries: From Bacterial Swarming to Effective Equilibrium.” Invited talk, *Brandeis Mathematical Biology Seminar* Mar. 28, 2024

Additional Oral Presentations

- APS Global Physics Summit, Denver 2026
- IHES Summer School, Bures-sur-Yvette 2025
- APS Global Physics Summit, Anaheim 2025
- APS New England Section Fall Meeting 2024
- Rosenfeld Symposium, Yale University 2019
- STARS Research Symposium, Yale University 2018

Poster Presentations

- International Conference on Statistical Physics, Florence 2025
- Boulder School for Condensed Matter Physics, Boulder 2024
- Beg Rohu Summer School, Beg Rohu 2023
- “Microtubules: From Atoms to Complex Systems,” EMBL meeting, Heidelberg 2022
- Biophysical Society Annual Meeting, San Francisco 2022

TEACHING

- Teaching assistant, Systems Biology (MIT 8.591) Fall 2026
- Recitation instructor, Introduction to Stochastic Dynamics (MIT 8.308) Spring 2025
- Recitation instructor, Graduate Statistical Mechanics (MIT 8.333) Fall 2023
- Teaching assistant, General University Physics (Yale PHYS 180–181) Fall 2019, Spring 2020

LEADERSHIP AND SERVICE

- President, Yale Arab Students Association 2019–2020
- Patient aide, Smilow Cancer Hospital 2018–2019
- Blood-pressure screening volunteer, Yale HAPPY 2018–2020

SKILLS AND EXPERTISE

- *Theoretical expertise:* stochastic processes, statistical field theory, disordered systems, and Monte Carlo methods.
- *Programming languages:* extensive experience in Julia and MATLAB; working knowledge of C and Python.
- *Laboratory skills:* single-molecule microscopy, microfluidics, protein biochemistry, molecular biology, and cell culture.
- *Languages:* native proficiency in English and Arabic.